

CoFlex-VTLA: Spatiotemporal Tactile Grounding for Adaptive Directional Compliance in Contact-Rich Manipulation

Yijie Luo^{1,2*}, Guanghui Qin^{1*}, Wei Wang^{1,2}, Xingyu Li^{1,2}, Zhipeng Wang^{1,2} *Member, IEEE*,
Yanmin Zhou^{1,2†} *Member, IEEE*, Bin He^{1,2} *Senior Member, IEEE*

¹Tongji University, ²Shanghai Research Institute for Intelligent Autonomous Systems

* Equal Contribution, †Corresponding Author

Abstract—Vision-Language-Action (VLA) models generalize well across manipulation tasks, but their kinematic actions do not specify how motions should be physically executed during physical interaction. Appropriate mechanical response depends not only on the evolving interaction, but also on task intent, task-space direction, and displacement polarity. We present CoFlex-VTLA, a Vision-Tactile-Language-Action (VTLA) framework that jointly predicts reference motion and task-conditioned directional compliance. Its Hierarchical Spatio-Temporal Tactile Modulation (HiST-TM) models spatial correlations in distributed tactile measurements and aggregates multi-rate histories to represent the evolving physical interaction. Fused with the contextual VLA representation, the policy jointly predicts reference action chunks and Asymmetric Directional Compliance Fields (ADCF), enabling direction- and polarity-dependent compliance, which is realized through high-rate tactile-guided compliant control. Across six real-world manipulation tasks with heterogeneous physical interaction requirements, CoFlex-VTLA achieves a 70.0% overall task-level success rate, outperforming the strongest baseline by 27.5%. It also improves contact-sensitive outcome consistency and force regulation, while generalizing to unseen watch-glass geometries and sizes without fine-tuning. These results show that jointly reasoning over task context and spatiotemporal physical interaction enables more reliable adaptation of how robot motions are physically executed. Our project page is available at <https://ultralyj.github.io/CoFlexVTLA>.

I. INTRODUCTION

Vision-Language-Action (VLA) models have demonstrated strong performance and promising generalization across diverse robotic manipulation tasks [1], [2], [3]. However, a kinematically appropriate motion does not fully specify how it should be physically executed [4]. In contact-rich manipulation, the same nominal motion can require markedly different mechanical responses as material properties, task-space constraints, and interaction conditions change [5], [6]. A robot may need to track accurately along one task-space direction while yielding along another, or progressively change its resistance as an interaction evolves from free motion to contact, loading, sustained interaction, and release. Because environmental contacts are often unilateral, even opposite displacement directions along the same Cartesian

This work was supported in part by the National Key Research and Development Program of China No. 2024YFB4709800, and in part by the National Natural Science Foundation of China No. 62473294, 62403363, 62403366. The corresponding author is Yanmin Zhou. (e-mail: yanmin.zhou@tongji.edu.cn)

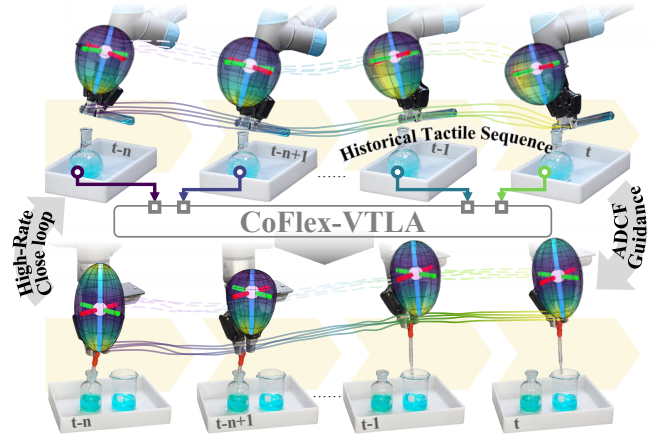


Fig. 1. CoFlex-VTLA adapts how VLA-generated motions are physically executed. Temporal tactile context grounds the evolving contact state, while adaptive directional compliance determines how strongly the robot resists or yields along each task-space direction.

axis can demand different responses [7]. Successful contact-rich manipulation therefore requires a VLA policy to determine not only *where to move*, but also *how strongly to resist or yield while moving*. This requirement exposes a central challenge: *how to adapt the robot's mechanical response to the evolving physical interaction*.

Recent VLA models have begun to incorporate tactile or force feedback to improve contact awareness and action prediction [6], [8], [9]. However, tactile feedback does not resolve how compliance should be regulated by itself. Existing approaches primarily use tactile information to refine motion or force actions, while compliance is often fixed, represented coarsely, or delegated to downstream controllers [10], [11], [12]. Although such controllers can execute compliant behavior in response to local motion or force errors, they do not determine what mechanical response is appropriate for the current task stage [13]. That response depends on the manipulation objective and can vary with interaction progress, task-space direction, and displacement polarity. Compliance, therefore, cannot be treated solely as a task-agnostic low-level reaction to contact. Rather, the VLA policy should specify the task-dependent mechanical behavior to be realized by the downstream controller.

Determining such compliance further requires a representation of how the physical interaction is evolving, rather than only its instantaneous contact state [14]. Similar instantaneous tactile observations can arise during contact establishment, increasing load, sustained contact, or release, while implying different requirements for resistance and yielding. The instantaneous tactile observation therefore does not generally constitute a sufficient Markov state for compliance regulation: its mechanical meaning depends on the preceding evolution of contact. Temporal tactile feedback provides this missing context by revealing how local interaction signals evolve over time rather than only their current magnitude [12], [9]. Moreover, similar aggregate contact signals can arise from different spatial contact distributions, which may impose different task-space constraints [15]. The spatial distribution of contact and its temporal evolution therefore jointly characterize the interaction state relevant to mechanical response. Compliance regulation should consequently be conditioned on spatiotemporal tactile information together with task context, enabling the VLA policy to explicitly specify task- and interaction-dependent mechanical responses across task-space directions and displacement polarities.

To realize this formulation, we introduce **CoFlex-VTLA**, a Vision-Tactile-Language-Action (VTLA) framework that jointly predicts reference motion and task-conditioned directional compliance. **Hierarchical Spatio-Temporal Tactile Modulation (HiST-TM)** models spatial correlations in distributed tactile measurements and aggregates contact history at multiple temporal scales to encode the evolving interaction state. Fused with the contextual VLA representation, the action expert jointly predicts reference action chunks and **Asymmetric Directional Compliance Fields (ADCF)**, which parameterize polarity-dependent translational stiffness along the Cartesian axes. High-rate tactile-guided compliant control then executes the paired motion-compliance commands under continuously updated tactile feedback.

Our main contributions are summarized as follows:

- We formulate contact-rich VLA manipulation as jointly specifying desired motion and task-dependent mechanical response, extending VLA policies beyond purely kinematic action generation.
- We propose **CoFlex-VTLA**, integrating HiST-TM for spatiotemporal interaction grounding, ADCF for direction- and polarity-dependent compliance prediction, and high-rate VIC execution for tactile-guided compliant manipulation.
- We evaluate CoFlex-VTLA on six real-world contact-rich laboratory tasks with heterogeneous physical interaction requirements. It achieves a 70.0% overall task-level success rate, outperforming the strongest baseline by 27.5 percentage points, while improving contact-sensitive physical outcomes and generalizing to unseen object geometries and sizes without fine-tuning.

II. RELATED WORK

A. Policies for Physical Interaction

Vision-Language-Action (VLA) models leverage pre-trained vision-language representations to generate robot actions with broad semantic and task-level generalization [2], [1], [3]. However, most VLAs primarily predict kinematic commands, specifying what motion to perform without explicitly modeling how that motion should be mechanically executed during contact.

Recent tactile-augmented VLA/VTLA models incorporate physical feedback into multi-modal representations or action prediction [8], [16], [17], [18], while force-aware policies further exploit interaction-force information [6], [4]. However, these methods primarily use physical feedback to refine motion or force prediction, rather than to explicitly regulate task-dependent compliance. CoFlex-VTLA instead grounds physical interaction from spatiotemporal tactile feedback and jointly predicts reference motion and directional compliance.

B. Spatiotemporal Tactile Grounding

Tactile interaction is inherently spatiotemporal: contact information is distributed across sensing locations and evolves over time. Prior work has modeled tactile history through temporal fusion [17], [19], multi-rate or slow-fast architectures [12], [20], and efficient long-horizon sequence models [9]. Recent studies further capture spatial structure together with temporal evolution for robotic manipulation [21], [22].

These representations are mainly used to improve tactile perception or action prediction. CoFlex-VTLA instead uses spatially structured, multi-rate tactile history to ground the evolving physical interaction and condition both reference motion and directional compliance.

C. Adaptive Compliance for Contact-Rich Manipulation

Compliance control shapes a robot’s mechanical response to physical interaction through parameters such as stiffness and damping [13]. Classical methods separate reference motion from mechanical behavior and adapt compliance across task-space directions according to interaction requirements [23]. Learning-based approaches further infer task-dependent compliance from demonstrations or interaction context [5].

More recent foundation-model-based methods generate adaptive compliance parameters, hybrid force–position actions, or high-level compliance commands from semantic and physical context [24], [10], [15]. CoFlex-VTLA differs by coupling spatiotemporal tactile grounding with joint reference-motion and asymmetric directional-compliance prediction, allowing compliance to vary with task context, interaction evolution, task-space direction, and displacement polarity.

III. METHODOLOGY

We present **CoFlex-VTLA**, a Vision-Tactile-Language-Action (VTLA) framework that jointly predicts reference motion and task-conditioned directional compliance. As illustrated in Fig. 2, CoFlex-VTLA incorporates tactile feedback at two complementary time scales. At the policy level,

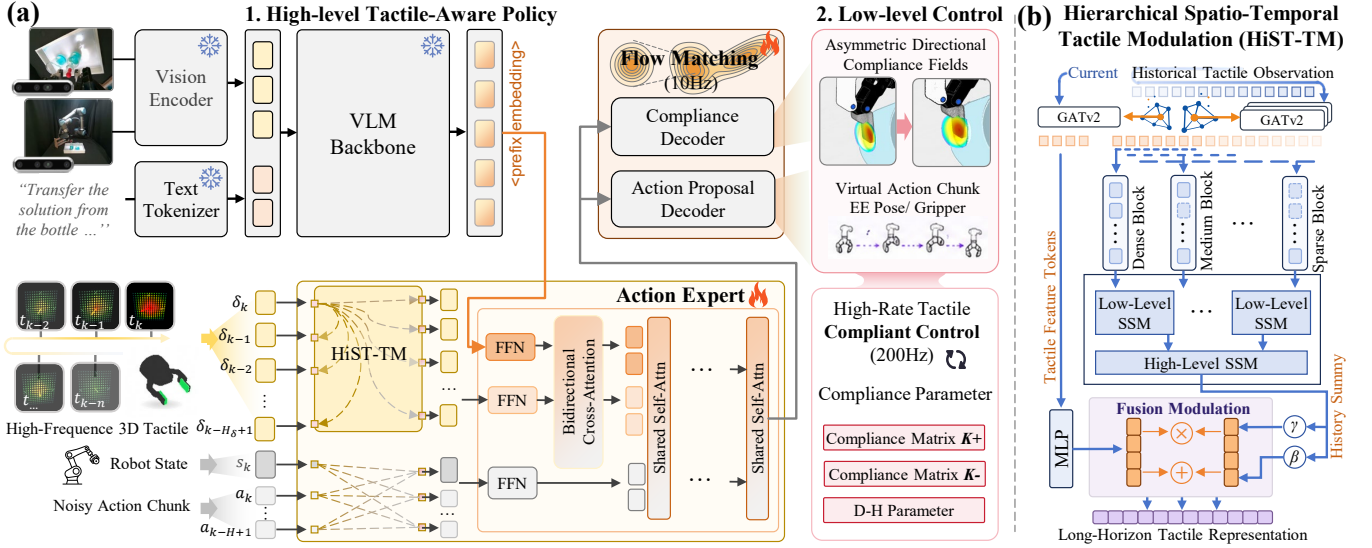


Fig. 2. **Overview of CoFlex-VTLA.** (a) The VLA backbone encodes language, vision, and robot state, while HiST-TM grounds the policy in the evolving local contact state using high-rate distributed tactile observations. After bidirectional fusion of the tactile and contextual VLA representations, the action expert jointly predicts reference action chunks and asymmetric directional compliance fields. The paired motion–compliance commands are realized through high-rate tactile-guided compliant control. (b) HiST-TM models spatial correlations among tactile measurements and hierarchically aggregates multi-rate contact histories to modulate the current tactile representation.

Hierarchical Spatio-Temporal Tactile Modulation (HiST-TM) encodes distributed tactile history into a spatiotemporal interaction representation, which is fused with the contextual VLA representation to jointly predict reference action chunks and **Asymmetric Directional Compliance Fields (ADCF)**. At the execution level, high-rate tactile feedback realizes the paired motion–compliance commands through compliant reference adaptation and control. We formulate the joint motion–compliance prediction problem and then introduce HiST-TM (Sec. III-B), ADCF-based motion–compliance prediction (Sec. III-C), and high-rate tactile-guided compliance execution (Sec. III-D).

A. Problem Formulation

Contact-rich manipulation couples high-level task reasoning with physical interaction dynamics that can evolve at a much faster time scale. Let k denote policy steps operating at frequency f_π . At each step, the policy observes a language instruction l , a visual observation I_k , and the robot state s_k , while tactile sensing and low-level control operate at higher frequencies $f_{\text{tac}}, f_{\text{ctrl}} \gg f_\pi$.

To represent the recent evolution of physical interaction, we associate each policy step with a tactile history

$$\mathcal{T}_k = (\mathbf{T}_{k,1}, \dots, \mathbf{T}_{k,L}), \quad \mathbf{T}_{k,j} \in \mathbb{R}^{N \times 3}, \quad (1)$$

where $\mathbf{T}_{k,j}$ contains three-axis force measurements from N distributed fingertip sensing points, and L denotes the tactile context length.

We formulate CoFlex-VTLA as a policy that jointly predicts

$$(\mathbf{A}_k, \mathbf{C}_k) = \pi_\theta(l, I_k, s_k, \mathcal{T}_k), \quad (2)$$

where $\mathbf{A}_k = (\mathbf{a}_{k,1}, \dots, \mathbf{a}_{k,H})$ and $\mathbf{C}_k = (\mathbf{c}_{k,1}, \dots, \mathbf{c}_{k,H})$ denote paired reference-action and directional-compliance

chunks over prediction horizon H . Together, they specify the desired motion and its task-conditioned mechanical response over the same prediction horizon.

B. Spatio-Temporal Interaction Grounding with HiST-TM

To represent the evolving physical interaction from distributed tactile feedback, we propose **Hierarchical Spatio-Temporal Tactile Modulation (HiST-TM)**, shown in Fig. 2(b). HiST-TM models spatial correlations within each tactile frame, summarizes interaction history over multiple temporal scales, and uses the resulting context to modulate the current tactile representation.

For each tactile frame $\mathbf{T} * k, j$, we model correlations among distributed sensing points using GATv2 and obtain a compact frame-level representation

$$\mathbf{z}_{k,j} = \text{Pool}(\text{GATv2}(\mathbf{T}_{k,j})). \quad (3)$$

The latest feature $\mathbf{z}_{k,L}$ represents the current spatial contact pattern, while preceding features capture how physical interaction has evolved over time.

A single temporal resolution may emphasize either rapid contact transitions or longer-term interaction trends. We therefore construct multi-rate historical sequences

$$\mathcal{Z}_k^{(s)} = [\mathbf{z}_{k,L-s}, \dots, \mathbf{z}_{k,L-Ms}], \quad s \in \mathcal{S}, \quad (4)$$

where $\mathcal{S}$ is a set of temporal sampling intervals and M is the number of observations retained in each stream, with $M_s \leq L - 1$. Small intervals preserve rapid local interaction changes, whereas larger intervals capture evolution over a broader temporal horizon.

The multi-rate histories are hierarchically aggregated using

two-stage S4D encoding:

$$\begin{aligned} \mathbf{u}_k^{(s)} &= \text{S4D}_1 \left(\mathbf{z}_k^{(s)} \right), \quad s \in \mathcal{S}, \\ \mathbf{h}_k &= \text{S4D}_2 \left(\text{Concat}_{s \in \mathcal{S}} \mathbf{u}_k^{(s)} \right). \end{aligned} \quad (5)$$

Here, $\mathbf{u}_k^{(s)}$ summarizes interaction evolution at one temporal scale, and $\mathbf{h}_k$ integrates these summaries into a multi-scale temporal context.

Rather than directly concatenating the historical context with the latest tactile feature, HiST-TM uses the preceding interaction to modulate the representation of the current observation. Specifically,

$$[\gamma_k, \beta_k] \text{MLP}(\mathbf{h} * k), \quad (6)$$

and

$$\tilde{\mathbf{z}}_k^t = (1 + \gamma_k) \odot \text{LN}(\mathbf{z}_{k,L}) + \beta_k. \quad (7)$$

This preserves the current spatial tactile observation while conditioning its representation on the preceding interaction, yielding a compact spatiotemporal representation of the current physical interaction.

To integrate this physical interaction with task-level visual and semantic context, let $\mathbf{Z}_k^{\text{vla}}$ denote the contextual representation produced by the VLA backbone and $\mathbf{Z}_k^t = \tilde{\mathbf{z}}_k^t$ the tactile representation. Information is exchanged through bidirectional cross-attention:

$$\tilde{\mathbf{Z}}_k^{\text{vla}} = \mathbf{Z}_k^{\text{vla}} + \text{MHA}(\mathbf{Z}_k^{\text{vla}}, \mathbf{Z}_k^t, \mathbf{Z}_k^t), \quad (8)$$

$$\tilde{\mathbf{Z}}_k^t = \mathbf{Z}_k^t + \text{MHA}(\mathbf{Z}_k^t, \tilde{\mathbf{Z}}_k^{\text{vla}}, \tilde{\mathbf{Z}}_k^{\text{vla}}). \quad (9)$$

The fused representations jointly encode task context and the evolving physical interaction, and condition the subsequent prediction of both reference motion and directional compliance.

C. Task-Conditioned Motion and Compliance Prediction

Given the fused task and interaction context, CoFlex-VTLA jointly predicts reference actions and their associated directional compliance. For each prediction step h , $\mathbf{a}_{k,h}$ specifies the reference action, while $\mathbf{c}_{k,h}$ parameterizes its task-conditioned mechanical response. We represent directional compliance through a stiffness parameterization termed the **Asymmetric Directional Compliance Field (ADCF)**:

$$\mathbf{c}_{k,h} = [k_x^+, k_x^-, k_y^+, k_y^-, k_z^+, k_z^-]_{k,h}, \quad (10)$$

where k_i^+ and k_i^- independently specify stiffness for positive and negative displacement polarities along translational axis $i \in \{x, y, z\}$. The predicted values are constrained to $[k_{\min}, k_{\max}]$. Rotational motion and gripper commands remain specified by the reference action.

Let

$$\mathbf{e} = \mathbf{x} - \mathbf{x}_r = [e_x, e_y, e_z]^\top \quad (11)$$

denote the translational displacement from the reference $\mathbf{x}_r$. For each axis, the applied stiffness is selected according to displacement polarity:

$$k_i(e_i) = \begin{cases} k_i^+, & e_i \geq 0, \\ k_i^-, & e_i < 0, \end{cases} \quad i \in \{x, y, z\}. \quad (12)$$

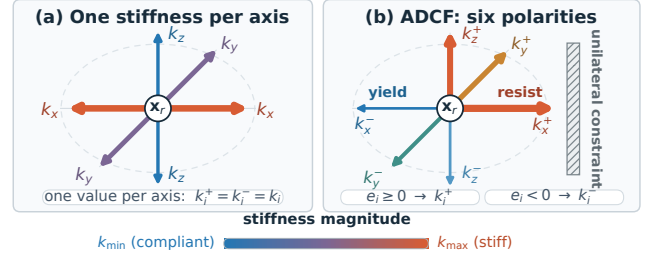


Fig. 3. **Asymmetric Directional Compliance Field (ADCF)**. (a) Conventional axis-wise compliance assigns a single stiffness value to each Cartesian axis, producing identical responses to positive and negative displacements. (b) ADCF independently parameterizes the six translational polarities. The displacement sign selects k_i^+ or k_i^- , enabling different mechanical responses on opposite sides of the same reference under direction-dependent or unilateral constraints. Color and line width encode stiffness magnitude from $k_{\min}$ (compliant) to $k_{\max}$ (stiff); $\mathbf{x}_r$ denotes the reference.

The resulting translational stiffness matrix is

$$\mathbf{K}_{k,h}(\mathbf{e}) = \text{diag}(k_x(e_x), k_y(e_y), k_z(e_z)). \quad (13)$$

ADCF, therefore, allows the policy to assign different mechanical responses to opposite displacement polarities independently along each translational axis, extending conventional axis-wise compliance beyond a symmetric stiffness parameterization.

Fig. 3 illustrates the asymmetric directional parameterization of ADCF. The same representation is used across tasks; at inference time, its values are predicted directly from the fused task and interaction context, without interaction-phase labels or pre-specified stiffness profiles.

D. High-Rate Tactile-Guided Compliance Execution

The policy specifies paired reference-motion and compliance commands at frequency f_π , while physical interaction can evolve substantially between consecutive policy updates. We therefore execute each predicted motion-compliance pair through high-rate tactile-guided reference adaptation and compliant control at $f_{\text{ctrl}} \gg f_\pi$.

At chunk element h , the reference $\mathbf{x}_r$ specifies the nominal motion, while ADCF determines the polarity-dependent stiffness $\mathbf{K}_{k,h}(\mathbf{e})$ that realizes the predicted directional compliance. During execution, high-rate tactile feedback adapts the nominal reference to ongoing interaction changes. Fingertip measurements are continuously aggregated into a Cartesian contact-force estimate $\mathbf{f}_\tau$ in the compliance-field frame, from which the virtual execution target is computed as

$$\mathbf{x}_v = \mathbf{x}_r - \mathbf{G}_\tau \mathbf{K}_{k,h}(\mathbf{e})^{-1} \text{clip}(\mathbf{f}_\tau - \mathbf{f}_{\text{ref}}), \quad (14)$$

where $\mathbf{G}_\tau$ is the tactile-feedback gain, $\mathbf{f}_{\text{ref}}$ is the task-specified force reference expressed in the compliance-field frame and held fixed throughout execution, and $\mathbf{x}_v$ is the virtual execution target. The force error is element-wise clipped to $[-f_{\max}, f_{\max}]$. Consequently, the same force error induces a larger virtual-target displacement along more compliant directions and a smaller displacement along less compliant directions.

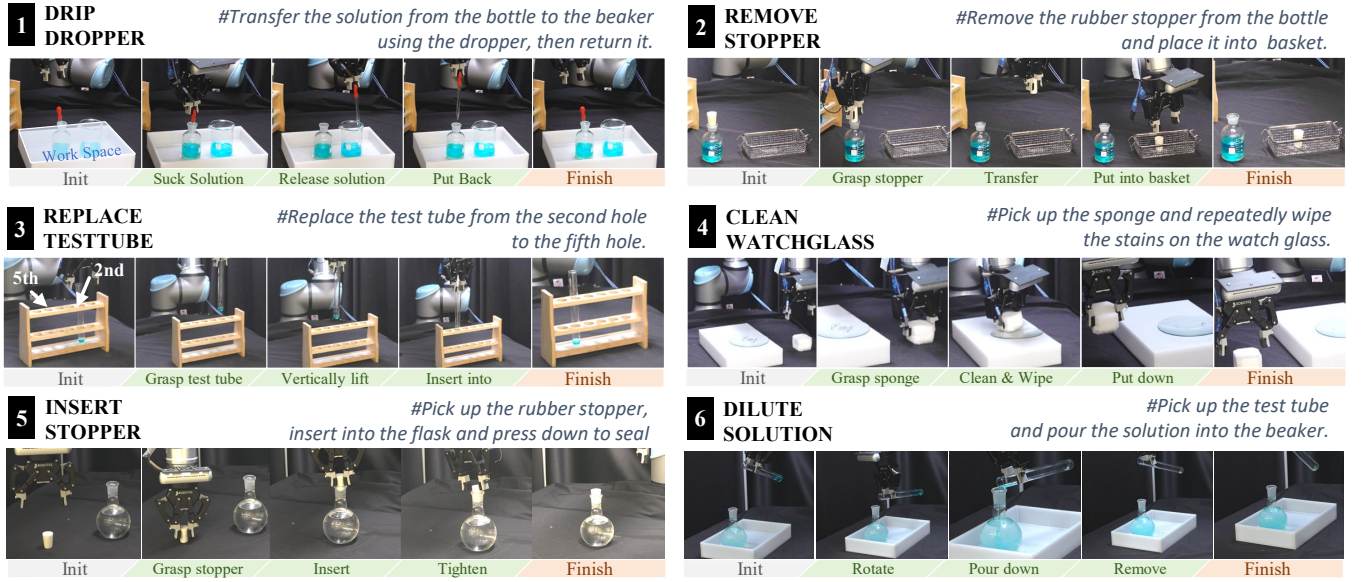


Fig. 4. Task categories, prompts and action primitives. The tasks are divided into three categories: grasper major (Drip Dropper, Remove Stopper), wrench major (Replace Test Tube, Dilute Solution), and twist major (Clean Watch Glass, Insert Stopper). The action primitives are designed to facilitate the execution of these tasks.

The resulting compliance behavior is realized by the following Cartesian stiffness–damping control law:

$$\mathbf{F}_c = \mathbf{K}_{k,h}(\mathbf{e})(\mathbf{x}_v - \mathbf{x}) + \mathbf{D}(\dot{\mathbf{x}}_v - \dot{\mathbf{x}}), \quad (15)$$

where $\mathbf{D}$ is the damping matrix.

Overall, CoFlex-VTLA separates policy-level compliance prediction from high-rate tactile-guided execution, enabling fast interaction adaptation without requiring high-frequency VTLA inference.

IV. EXPERIMENTS

Our experiments evaluate the complete CoFlex-VTLA interaction loop beyond task success. Specifically, we examine whether spatiotemporal tactile grounding and adaptive directional compliance improve contact-rich execution and physical regulation, how the proposed components contribute to these gains, and whether the learned interaction strategy generalizes beyond the training conditions. We address four questions:

Q1: Overall effectiveness. Does CoFlex-VTLA improve execution success across contact-rich tasks with heterogeneous physical interaction requirements?

Q2: Asymmetric Directional Compliance Adaptation. Does task- and contact-conditioned motion–compliance prediction produce appropriate and consistent mechanical responses across task-space directions and displacement polarities as interaction conditions evolve?

Q3: Component necessity. How do spatiotemporal tactile grounding, asymmetric directional compliance, and high-rate tactile feedback individually contribute to contact-rich execution?

Q4: Generalization. Does the learned task- and contact-conditioned interaction strategy transfer to unseen object geometries and sizes without policy fine-tuning?

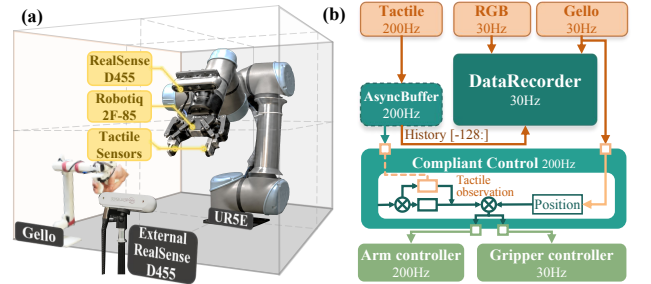


Fig. 5. **Experimental platform and multi-rate data pipeline.** (a) Real-world manipulation platform with a GELLO teleoperator for demonstration collection. (b) Asynchronous sensing and control pipeline integrating visual observations, distributed tactile feedback, robot state, and high-rate compliant control.

A. Experimental Setup

Platform. As shown in Fig. 5(a), experiments use a UR5E arm with a Robotiq 2F-85 gripper, two fingertip-mounted magneto-tactile sensors (MC-M2020-Elite, Paxini), and two RealSense D455 cameras. Demonstrations are collected via GELLO teleoperation.

Task Design. We construct six chemistry-inspired laboratory tasks spanning heterogeneous interaction regimes (Fig. 4): Drip Dropper, Remove Stopper, Replace Test Tube, Dilute Solution, Clean Watch Glass, and Insert Stopper. Together, they cover deformable-object interaction, constrained extraction and insertion, sustained surface contact, and phase-varying task-space constraints.

For analysis, we group the tasks by their dominant source of contact sensitivity into grasper-dominant, wrench-dominant, and twist-dominant categories. Their stages span free motion, contact establishment, load-bearing interaction, constrained motion, and release, requiring the robot

TABLE I
TASK-LEVEL SUCCESS RATES ACROSS SIX CONTACT-RICH TASKS.

Task Classification		Grasper Dominant					Wrench Dominant				Twist Dominant				Avg. (%)
Tactile	Policy	Drip Dropper			Remove Stopper		Replace Test Tube		Dilute Solution		Clean Watch Glass		Insert Stopper		
		grasp	suck	drip	remove	put	lift	insert	grasp	pull	take	clean	move	insert	
w/o Tac.	DP[25]	12/20	5/20	0/20	16/20	16/20	8/20	0/20	9/20	0/20	15/20	5/20	20/20	14/20	29.2
	$\pi_{0.5}$ [3]	18/20	12/20	2/20	14/20	10/20	11/20	1/20	12/20	0/20	17/20	11/20	18/20	11/20	29.2
	WALL-X[26]	20/20	15/20	3/20	16/20	14/20	13/20	1/20	17/20	0/20	19/20	15/20	18/20	12/20	37.5
w/ Tac.	ACP[5]	5/20	2/20	0/20	19/20	19/20	17/20	13/20	12/20	2/20	14/20	5/20	18/20	14/20	44.2
	$\pi_{0.5}+\text{tac}$. [3]	3/20	0/20	0/20	7/20	5/20	2/20	0/20	11/20	1/20	12/20	0/20	0/20	0/20	5.0
	RDP[12]	17/20	14/20	5/20	13/20	9/20	15/20	5/20	17/20	5/20	19/20	16/20	12/20	11/20	42.5
	Ours	19/20	15/20	11/20	20/20	20/20	20/20	9/20	17/20	13/20	18/20	15/20	17/20	16/20	70.0

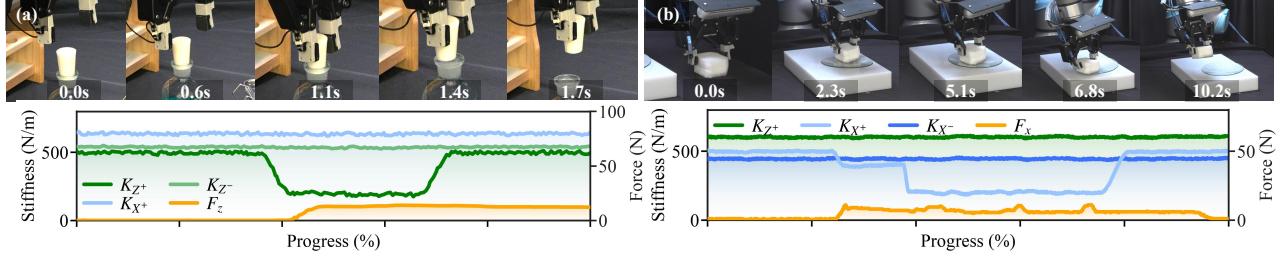


Fig. 6. Representative force and stiffness evolution under adaptive directional compliance.(a) stopper removal. (b) watch-glass cleaning. The plots show the contact force and predicted stiffness along a representative task-relevant axis across interaction stages, together with the corresponding execution snapshots.

to ground the evolving physical interaction and adapt its mechanical response. The benchmark therefore evaluates both spatiotemporal tactile grounding and task- and contact-conditioned compliance adaptation.

Each task comprises multiple stages with distinct physical requirements. We collect 200 demonstrations per task, totaling 527K frames, and use task-level evaluation to localize failures around contact transitions.

Baselines. We compare CoFlex-VTLA with tactile-free and tactile-enabled baselines. The tactile-free group includes Diffusion Policy (DP) [25], $\pi_{0.5}$ [3], and WALL-X [26], providing strong visuomotor and VLA references without local physical feedback.

The tactile-enabled group includes ACP [5], RDP [12], and $\pi_{0.5}+\text{tac}$. ACP learns compliance from wrench feedback, RDP employs slow-fast visual-tactile control, and $\pi_{0.5}+\text{tac}$ directly concatenates tactile observations with robot state. Together, these baselines provide complementary comparisons across tactile sensing, temporal tactile modeling, and explicit compliance adaptation.

Evaluation Protocol. All methods use the same demonstrations and training epochs and are evaluated under identical task definitions, success criteria, and real-world conditions, with their respective observation settings. For ACP, fingertip measurements are converted to wrench inputs.

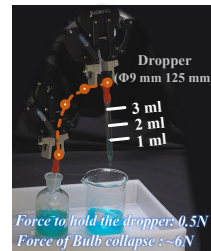
Physical regulation through volume consistency, force-profile agreement, and direction- and polarity-dependent stiffness evolution; component contributions through ablations of spatiotemporal tactile grounding, compliance representation, and high-rate tactile feedback; and generalization to unseen geometries and sizes without fine-tuning. These metrics distinguish task completion from physically appro-

priate contact behavior.

Implementation Details. CoFlex-VTLA uses the $\pi_{0.5}$ backbone. Tactile sensing and compliant control operate at 200 Hz, visual observations and demonstrations at 30 Hz, and policy inference at 10 Hz (Fig. 5(b)). HiST-TM uses a 128-frame tactile context, and pose regression adopts the continuous 6D rotation representation [27]. The model is trained for 20k optimization steps. Directional-compliance supervision is constructed through a one-time calibration at the interaction-phase level. For each task phase, a six-dimensional polarity-dependent stiffness target is specified according to its task-space constraints and shared across all demonstrations of that phase. These targets supervise the compliance branch jointly with reference-action prediction. During inference, the policy predicts ADCF directly from the grounded multimodal context, without phase labels or online stiffness tuning.

TABLE II
DISPENSED-VOLUME STATISTICS FOR DIFFERENT POLICIES.

Policy	Avg. $\pm$ Std.	In Range
DP[25]	0.0 $\pm$ 0.0	0%
$\pi_{0.5}$ [3]	0.2 $\pm$ 0.9	20%
WALL-X[26]	0.8 $\pm$ 1.5	0%
ACP[5]	1.1 $\pm$ 2.4	0%
$\pi_{0.5}+\text{tac}$. [3]	1.5 $\pm$ 1.1	0%
RDP[12]	1.8 $\pm$ 1.6	0%
Ours	1.9 $\pm$ 0.6	60%



B. Evaluation Results

Q1: Overall effectiveness.

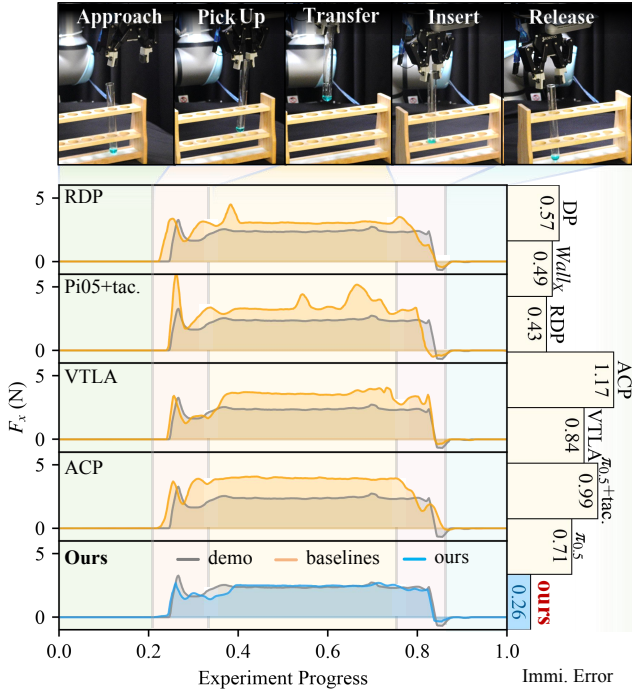


Fig. 7. Comparison between reference and policy-generated force profiles during test-tube manipulation.

We first examine whether CoFlex-VTLA improves reliability across heterogeneous contact regimes rather than a particular manipulation primitive. As shown in Table I, CoFlex-VTLA achieves the highest overall task-level success rate of **70.0%**, exceeding the strongest baseline, RDP (42.5%), by 27.5%. The gains span grasper-, wrench-, and twist-dominant tasks and are largest in stages involving sustained or constrained contact, where successful execution depends on adapting to evolving physical conditions rather than following a plausible trajectory alone.

The $\pi_{0.5}+\text{tac.}$ comparison further clarifies the source of this improvement. Despite direct access to tactile measurements, it achieves only 5.0% and performs worse than the original $\pi_{0.5}$. Thus, tactile input alone is insufficient and may even hinder the policy when not appropriately integrated. CoFlex-VTLA benefits from using tactile history to ground the interaction state and coupling that state with an adaptive mechanical response.

Q2: Asymmetric Directional Compliance Adaptation.

Task success alone cannot determine whether contact is executed with an appropriate mechanical response. We therefore analyze force–stiffness trajectories and contact-sensitive outcomes.

Adaptive force–stiffness evolution. Fig. 6 compares stopper removal, characterized by constrained engagement and release, with watch-glass cleaning, which requires sustained tangential contact. In both tasks, stiffness evolves with interaction progress and differs across task-relevant Cartesian directions. Moreover, opposite displacement polarities along the same axis can exhibit distinct stiffness profiles, reflecting the unilateral mechanical constraints captured by

ADCF. These patterns show that CoFlex-VTLA adapts its mechanical response jointly to interaction state, direction, and displacement polarity rather than applying fixed or symmetric compliance.

Dropper regulation. Drip Dropper evaluates the consistency of deformation-sensitive liquid dispensing. CoFlex-VTLA dispenses 1.9 ± 0.6 [unit], with 60% of trials within the target range (Table II). While some baselines achieve comparable mean volumes, their larger variance or lower in-range rates indicate less consistent physical regulation.

Test-tube manipulation. Replace Test Tube transitions from load-bearing transfer to constrained insertion. As shown in Fig. 7, CoFlex-VTLA follows the reference force profile more consistently through this transition, indicating adaptation to changing physical constraints rather than maintaining a uniform mechanical response throughout the trajectory.

Qualitative failure analysis. Fig. 8 shows that baseline failures frequently occur near contact transitions or during sustained interaction, including liquid handling, stopper gripping, surface cleaning, and object return. These cases are consistent with inadequate interaction grounding or mechanical adaptation, although they do not establish causality.

C. Ablations and Analysis

Q3: Component necessity.

We isolate the roles of compliance representation and temporal tactile grounding on three representative contact-dominant stages (Table III). The full model achieves **71.7%**. Replacing ADCF with symmetric compliance reduces performance to 63.3%, showing the importance of polarity-dependent mechanical responses. Fixed compliance further reduces performance to 60.0%, while removing explicit compliance prediction leads to a larger drop to 28.3%, confirming the benefit of adapting mechanical response to the evolving interaction.

Temporal grounding is likewise important. Using only single-rate tactile history reduces performance to 53.3%, while relying solely on the current tactile observation further decreases it to 35.0%. These results indicate that contact history is necessary for grounding interaction evolution, and that multi-rate temporal context provides additional benefit over a single temporal scale.

TABLE III

ABLATION STUDY ON REPRESENTATIVE CONTACT-DOMINANT STAGES.

Variant	Test-Tube Insertion	Stopper Removal	Stopper Insertion	Avg.
CoFlex-VTLA	9/20	20/20	14/20	71.7%
Sym. Compliance	6/20	20/20	12/20	63.3%
Fixed Compliance	6/20	18/20	12/20	60.0%
No Compliance	1/20	10/20	6/20	28.3%
Single-Rate History	5/20	16/20	11/20	53.3%
Current Tactile Only	3/20	9/20	9/20	35.0%

Q4: Generalization to unseen geometry.

We evaluate watch-glass cleaning on unseen shapes and sizes, which alter the geometry of sustained contact while

preserving the task objective. This setting tests whether the learned interaction strategy transfers beyond the training geometry rather than merely reproducing a familiar trajectory.

As shown in Table IV, CoFlex-VTLA performs best across all four conditions, achieving 5/5 and 3/5 successes on medium and large square glasses and 4/5 on both round variants. The large-round condition provides the clearest distinction: CoFlex-VTLA succeeds in 4/5 trials, whereas all baselines fail. Within this task, the results indicate that contact-conditioned behavior can adapt to changes in surface shape and scale without fine-tuning.

TABLE IV
SUCCESS RATES FOR DIFFERENT OBJECT SHAPES AND SIZES.

Policy	Square		Round	
	Medium	Large	Medium	Large
DP[25]	2/5	0/5	0/5	0/5
$\pi_{0.5}$ [3]	3/5	0/5	0/5	0/5
WALL-X[26]	4/5	1/5	1/5	0/5
ACP[5]	1/5	0/5	1/5	0/5
$\pi_{0.5} + \text{tac}$ [3]	0/5	0/5	0/5	0/5
RDP[12]	4/5	2/5	2/5	0/5
Ours	5/5	3/5	4/5	4/5

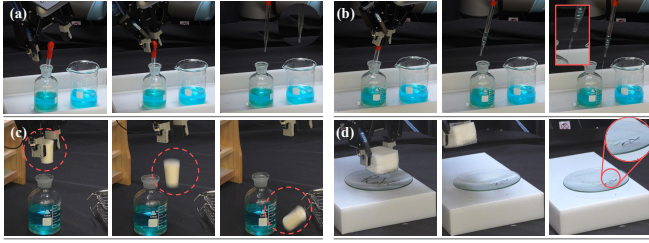


Fig. 8. Representative failure cases of baseline policies across three contact-rich manipulation tasks. (a) Failure to uptake liquid. (b) Premature liquid release. (c) Insufficient stopper grip. (d) Incomplete surface cleaning.

V. CONCLUSION

We presented **CoFlex-VTLA**, a Vision-Tactile-Language-Action (VTLA) framework that extends robot action prediction from reference motion to task-conditioned directional compliance. CoFlex-VTLA uses HiST-TM to represent the evolving physical interaction from spatially distributed, multi-rate tactile history, and jointly predicts reference action chunks and asymmetric directional compliance through ADCF. High-rate tactile-guided compliant control further realizes the predicted motion-compliance commands during execution. Across six real-world manipulation tasks with heterogeneous physical interaction requirements, CoFlex-VTLA achieves a 70.0% overall task-level success rate, outperforming the strongest baseline by 27.5%, while improving contact-sensitive physical outcomes and generalizing to unseen object geometries and sizes without fine-tuning. These results demonstrate the value of jointly reasoning about task intent and evolving physical interaction to determine not only how the robot should move, but also how compliantly that motion should be executed.

REFERENCES

- [1] M. J. Kim, K. Pertsch, S. Karamcheti *et al.*, “Openvla: An open-source vision-language-action model,” in *The 8th Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, P. Agrawal, O. Kroemer, and W. Burgard, Eds., vol. 270, 06–09 Nov 2025, pp. 2679–2713.
- [2] B. Zitkovich, T. Yu, S. Xu *et al.*, “Rt-2: Vision-language-action models transfer web knowledge to robotic control,” in *Conference on Robot Learning*. PMLR, 2023, pp. 2165–2183.
- [3] K. Black, N. Brown, J. Darvinian *et al.*, “ $\pi_{0.5}$: a vision-language-action model with open-world generalization,” in *9th Annual Conference on Robot Learning*, 2025.
- [4] H. Fang, S. Tang, M. Mei *et al.*, “Force policy: Learning hybrid force-position control policy under interaction frame for contact-rich manipulation,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2026.
- [5] Y. Hou, Z. Liu, C. Chi *et al.*, “Adaptive compliance policy: Learning approximate compliance for diffusion guided control,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 4829–4836.
- [6] J. Yu, H. Liu, Q. Yu *et al.*, “Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation,” *arXiv preprint arXiv:2505.22159*, 2025.
- [7] M. Zhu, D. Gong, Y. Zhao *et al.*, “Compliant force control for robots: A survey,” *Mathematics*, vol. 13, no. 13, 2025.
- [8] J. Huang, S. Wang, F. Lin *et al.*, “Tactile-vla: Unlocking vision-language action model’s physical knowledge for tactile generalization,” *arXiv preprint arXiv:2507.09160*, 2025.
- [9] Z. Wang, Y. Wang, M. Ren *et al.*, “Tacmamba: A tactile history compression adapter bridging fast,” *arXiv preprint arXiv:2603.01700*, 2026.
- [10] Y. Li, Zhaxizhuoma, H. Jiang *et al.*, “Forcevla2: Unleashing hybrid force-position control with force awareness for contact-rich manipulation,” *arXiv preprint arXiv:2603.15169*, 2026.
- [11] Z. Xu, Y. Wang, B. Abbatematteo *et al.*, “Contact-grounded policy: Dexterous visuotactile policy with generative contact grounding,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2026.
- [12] H. Xue, J. Ren, W. Chen *et al.*, “Reactive diffusion policy: Slow-fast visual-tactile policy learning for contact-rich manipulation,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2025.
- [13] F. J. Abu-Dakka and M. Saveriano, “Variable impedance control and learning—a review,” *Frontiers in Robotics and AI*, vol. Volume 7 - 2020, 2020.
- [14] J. Bi, K. Y. Ma, C. Hao *et al.*, “Vla-touch: Enhancing vision-language-action models with dual-level tactile feedback,” *arXiv preprint arXiv:2507.17294*, 2025.
- [15] H. Cao, Z. Li, Z. You *et al.*, “Paco-vla: Passivity-shielded compliance prior for contact-rich vision-language-action manipulation,” 2026.
- [16] P. Hao, C. Zhang, D. Li *et al.*, “Tla: Tactile-language-action model for contact-rich manipulation,” *arXiv preprint arXiv:2503.08548*, 2025.
- [17] C. Zhang, P. Hao, X. Cao *et al.*, “Vtla: Vision-tactile-language-action model with preference learning,” *arXiv preprint arXiv:2505.09577*, 2025.
- [18] Z. Cheng, Y. Zhang, W. Zhang *et al.*, “Omnivtla: Vision-tactile-language-action model with semantic-aligned tactile sensing,” *arXiv preprint arXiv:2508.08706*, 2025.
- [19] M. Ruan, L. Zhou, H. Li *et al.*, “Retac-act: A state-gated vision-tactile fusion transformer for precision assembly,” 2026.
- [20] X. Kang, T. Tian, S.-W. Lee *et al.*, “Learning force-regulated manipulation with a low-cost tactile-force-controlled gripper,” 2026.
- [21] Y. Ning, F. Zhao, and Q. Liu, “Touchwgnn: spatio-temporal tactile perception for multimodal dexterous manipulation,” *Frontiers in Robotics and AI*, vol. 13, p. 1791424, 2026.
- [22] C. Higuera, A. Sharma, T. Fan *et al.*, “Tactile beyond pixels: Multi-sensory touch representations for robot manipulation,” in *Conference on Robot Learning*. PMLR, 2025, pp. 105–123.
- [23] A. Calanca, R. Muradore, and P. Fiorini, “A review of algorithms for compliant control of stiff and fixed-compliance robots,” *IEEE/ASME Transactions on Mechatronics*, vol. 21, no. 2, pp. 613–624, 2016.
- [24] H. Zhang, W.-H. Huang, Q. Tong *et al.*, “Compliantvla-adaptor: Vlm-guided variable impedance action for safe contact-rich manipulation,” 2026.
- [25] C. Chi, S. Feng, Y. Du *et al.*, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

- [26] A. Zhai, B. Liu, B. Fang *et al.*, “Igniting vlms toward the embodied space,” *arXiv preprint arXiv:2509.11766*, 2025.
- [27] Y. Zhou, C. Barnes, J. Lu *et al.*, “On the continuity of rotation representations in neural networks,” in *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 5738–5746.